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EDUCATION

University of Texas at Dallas

Richardson, TX — August 2025 - Present

B.Tech in Computer Science, Honors Collegium V, AES Scholar

TECHNICAL SKILLS

Languages and Frameworks: Python, C++ , Java, Qiskit, JavaScript, SQL, React.js, Node.js, Flutter, Flask, HTML, CSS

Technologies: Qiskit, Pennylane, Android Studio, Scikit-learn, Auth0, Docker, Git, MongoDB, Google Cloud, Gemini API, ElasticSearch, Tableau, Postman, Figma, Dash, Plotly, n8n, openrouter

Specializations: AI APIs integration, Natural Language Processing (NLP), Deep Learning, Quantum, Data Analytics, Post-Quantum Cryptography

EXPERIENCE

IBM Quantum - Qiskit Advocate

Remote — September 2025 - Present

- Selected as one of over 540 global Qiskit Advocates contributing to quantum software education and open-source development across 30+ countries.
- Accepted into QAMP Research Mentorship Program with a 5% acceptance rate, collaborating on Q-UCSpec research integrating TDDFT spectroscopy with Quantum Machine Learning.
- Invited to IBM Quantum Developer Conference 2025, engaging with 200+ researchers on superconducting qubits, error mitigation, and quantum software tooling.

Nebula Labs (UTD) - Software Engineer

Richardson, TX — September 2025 - Present

- Collaborating on 2 open-source platforms serving 10,000+ UTD students: *UTD Trendz* (professor rating platform) and *UTD Notebook* (academic note-sharing platform).
- Contributing to backend API development using React.js, Node.js, and MongoDB; exploring GoLang integrations.
- Working with 30+ developers in cross-functional meetings following agile workflows, code reviews, and iterative releases.

Association for Computing Machinery (ACM) UTD Chapter - Research

Richardson, TX — August 2025 - November 2025

- Evaluated RLCE decoding behavior across 15 million test cases, identifying failure regimes beyond t-error correction.
- Designed a recurrent neural decoding approach achieving a 30x improvement in successful decoding beyond classical limits.
- Awarded 1st Place at the Fall 2025 ACM Research Symposium.

University of Texas at Dallas - Quantum Lab - Research Intern

Richardson, TX — June 2024 - August 2024

- Conducted 10+ weeks of quantum software research under Dr. Basu and Dr. Koludrubetz using Qiskit 1.x and ZX-calculus.
- Identified and resolved circuit verification bugs in pyZX by adding missing gate configurations; handled QASM parser issues with custom gates.
- Coordinated weekly lab meetings with a team of 5+ researchers to test solutions and present findings.

PROJECTS

HarmoniQ - AI Network Health Platform

November 2025

- Built an AI system combining sentiment analysis and network analytics to predict network outages; monitored real-time reddit posts and trained ML models achieving 87-92% accuracy.
- Awarded 3rd Place at HackUTD 2025, the largest 24-hour hackathon in North America, for the T-Mobile Challenge; recognized for Best Use of Auth0 by MLH.

ChatCycle - AI Knowledge Platform

August 2025 - November 2025

- Developed a platform for posting AI chats that helps reduce repeated questions in student communities; integrated GPT API to chat within the app.
- Designed concise AI responses and semantic search; reduced duplicate queries by approximately 40%.

SmartCartAI - Intelligent Shopping Agent

February 2025 - August 2025

- Engineered an AI agent converting grocery lists of 15+ items into automated e-commerce workflows using NLP and browser-use open-source software automation.
- Integrated Flask with n8n pipelines; reduced manual ordering time by approximately 60%.

StreamSign - AI Accessibility Tool

June 2024 - August 2024

- Built an ML pipeline translating video content into Indian Sign Language animations to improve accessibility for deaf users.
- Applied NLP preprocessing and evaluation techniques; achieved over 90% translation accuracy on validation samples.

PROGRAMS AND CERTIFICATIONS

- **TRAIN AI Research Program (The Coding School)** - Fulfilled an year long coursework in machine learning, NLP, and deep learning. Followed by an independent research program, chosen as 1 of 10 students from 2,000+ applicants, developed *StreamSign* as the final applied research project.
- **New York Academy of Sciences** - Young AI Researcher in a program with a 6–10% global acceptance rate; worked with an international team to build an AI-based waste management model.
- **Qiskit Global Summer School (IBM)** - Completed programs in 2021, 2023, and 2025 with a 100% final score in 2025.
- **Womanium Quantum Program** - Selected for 2023 and 2025 cohorts; 8+ weeks of advanced training in quantum algorithms, quantum machine learning, and applied research projects including PDE solvers and portfolio optimization.
- **QubitxQubit** - Completed a 2-semester foundational quantum computing program covering quantum gates, algorithms, and 20+ hands-on Qiskit labs in collaboration with IBM Quantum (2020- 2021).

AWARDS AND ACHIEVEMENTS

- 3rd Place - HackUTD 2025 (T-Mobile Challenge).
- IBM-Certified Quantum Developer (Qiskit v0.2x).
- 1st Place - Yale Quantum Hackathon 2025 (BlueQubit Virtual Challenge).
- Global Gold Medalist - ICPC AlgoQueen 2023 (Quantum and Competitive Programming).
- Advanced Badge Recipient and Top Scorer - IBM Quantum Challenges (2021, 2023, 2024, 2025).
- UIL Districts Silver Medalist - Competitive Programming
- Indian National Olympiad of Informatics Qualifier through Zonal Computing Olympiad 2022.